

Exercises

1. Explain the minimax objective function of a GAN: 1) What does each term present? 2) Why is it formulated as a game between two networks? 3) What would happen if the discriminator becomes too strong too quickly?
2. Why can GAN training suffer from vanishing gradients early in training?
3. Design a simple GAN for generating 28×28 grayscale images (like MNIST). What should the generator architecture look like (layers, activations)? What should the discriminator architecture look like? Why are BatchNorm and LeakyReLU commonly used?
4. Implement a basic GAN training loop: train on MNIST or Fashion-MNIST, Alternate updates between discriminator and generator, Use Binary Cross-Entropy (BCE) loss, ample noise vector $z \sim \mathcal{N}(0,1)$.
5. Consider the sentence: *"The animal didn't cross the street because it was too tired."* What does "it" refer to? Explain how self-attention helps the model resolve this.
6. How many scalar dot products are computed per head in the attention matrix?
7. Implement the scaled dot-product attention from scratch using numpy and test it with random matrices of shape (4,8). Verify that attention weights sum to 1 on each row.
8. Extend the previous to implement MHA.